DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-D32: Retail sales [20 marks]

Scenario: You are the analyst for a retail chain. The series sales (Sales (lakh rupees)) is recorded quarterly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 4 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(20703)
m <- 4                      # observations per cycle (quarterly data)
n <- m * 12                 # 12 full cycles
t <- 1:n
trend <- 130.6 + 1.545 * t
sales <- round(trend + 26.1 * cos(2 * pi * (t - 3) / m) + rnorm(n, 0, 6.4), 1)
x <- ts(sales, frequency = m, start = c(2016, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(sales ~ 1:n) and report the slope per quarter). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 4 seasonal values). Name the peak quarter and the trough quarter with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 12 cycles is the highest value in the peak quarter from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(sales ~ 1:n))) and compute acf(d, lag.max = 4, plot = FALSE) using observation lags. Report the autocorrelation at lag 4 and at lag 2, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of sales in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per quarter [2]. Write one sentence of advice to management based on your findings [2]. [6]

Remember: *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____

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Series SP-D32 : Retail sales [20 marks]

1) Series plot and basic information [3]



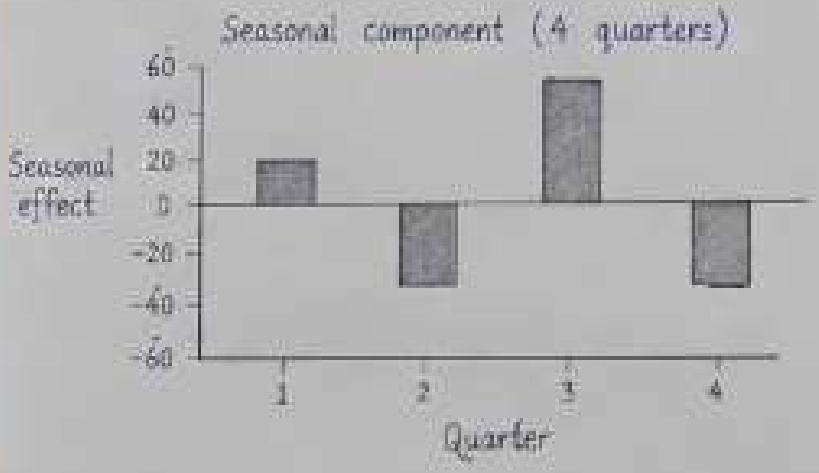
- n = 48 (observations)
- Cycle length = 4 (quarters)
- Number of full cycles = 12
- Overall trend is rising.
- The series shows a repeating seasonal pattern with increasing level.

Estimated slope using linear model : sales ~ t (from R output)

Slope = 1.545 lakh rupees per quarter.

2) Seasonal pattern [4]

R code : `dc <- decompose(x)`
`seasonal <- dc$seasonal[1:4]`



Quarter	Seasonal value (lakh Rs)
1	16.5
2	-28.7
3	52.1
4	-39.9

- Peak quarter = 3
- Peak value = 52.1 (lakh Rs)
- Trough quarter = 4
- Trough value = -39.9 (lakh Rs)
- Seasonal swing = 52.1 - (-39.9) = 92.0 (lakh Rs)

3) Seasonal plot (12 cycles) [3]

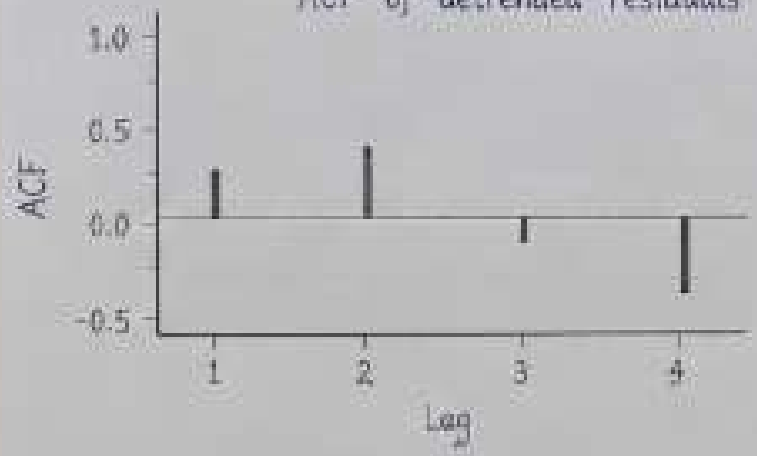


- The peak occurs at quarter 3 in all 12 cycles.
- Number of cycles with the same peak month as Task 2 = 12
- The seasonal pattern is consistent across the 12 cycles.

4) Autocorrelation analysis [4]

R code : `d <- resid(lm(sales ~ t))`
`acf(d, lag.max = 4, plot = FALSE)`

Lag	ACF value
1	0.068
2	0.412
3	-0.081
4	-0.356



Significance bound = $2/\sqrt{n} = 2/\sqrt{48} = 0.289$

- Lag 2 autocorrelation = 0.412
  $(|0.412| > 0.289) \rightarrow$ significant.
- Lag 4 autocorrelation = -0.356
  $(|0.356| > 0.289) \rightarrow$ significant
- The two values indicate the presence of seasonality (positive at lag 2 and negative at lag 4).

5) Additive or multiplicative? [6]

Cycle	Max (lakh Rs)	Min (lakh Rs)	Range (lakh Rs)
Cycle 1 (Q1-Q4)	162.5	98.3	64.2
Cycle 12 (Q45-Q48)	296.8	214.7	82.1

Ratio = last-cycle range / first-cycle range = 82.1 / 64.2 = 1.28

- Since the ratio is clearly above 1, the model is multiplicative (seasonal swings grow with the level).
- The seasonal variation increases as the sales level increases.

R code to fit model after removing seasonal component :

```
dc <- decompose(x)
seasonal <- dc$seasonal
adjusted <- sales - seasonal
trend <- lm(adjusted ~ t)
summary(trend)
```

Slope of adjusted series
= 1.486
Lakh rupees per quarter
(from R output)

6) Management advice (one sentence) [2]

Plan inventory and promotional activities by considering the 4-quarter seasonal cycle, with higher sales expected in quarter 3 and lower sales in quarter 4, and also account for the increasing trend when forecasting future demand.